TRIBHUVAN UNIVERSITY
FACULTY OF HUMANITIES AND SOCIAL SCIENCES

Library Management System

A Project Report

Submitted to:

Department of Computer Application

Madan Bhandari Memorial College

New baneshwor ,Kathmandu

In partial fulfillment of the requirements for the Bachelors in Computer Application

Submitted By:

Binita Kumari Khatri (6-2-432-120-2023)
Manisha Sapkota (6-2-432-130-2023)

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Under the supervision of

Mr. Phul Babu Jha

Tribhuvan University

Faculty of Humanities and Social Sciences

Madan Bhandari Memorial College

SUPERVISOR'S RECOMMENDATION

I hereby recommend that this project prepared under my supervision by Binita khatri and Manisha Sapkota entitled "Library Management System" in partial fulfillment of the requirements for the degree of Bachelor of Computer Application is recommended for the final evaluation.

……

SIGNATURE

Mr. Phul Babu Jha

SUPERVISOR

Madan Bhandari Memorial College (MBMC)

New Baneshwor, Kathmandu

Tribhuvan University

Faculty of Humanities and Social Sciences

Madan Bhandari Memorial College

LETTER OF APPROVAL

This is to certify that this project prepared by Binita Khatri and Manisha Sapkota entitled "**Library Management System**" in partial fulfillment of the requirements for the degree of Bachelor in Computer Application has been evaluated. In our opinion it is satisfactory in terms of scope and quality as a project for the required degree.

…… Mr. Phul Babu Jha Project Supervisor Madan Bhandari Memorial College New Baneshwor, Kathmandu	……. Naw Raj Joshi Coordinator Madan Bhandari Memorial College New Baneshwor, Kathmandu
…… Internal Examiner	…….. External Examiner

Abstract

The Library Management System (LMS) is a web-based application developed to automate and manage all major activities of a library through an online platform. The system is designed to replace manual record-keeping with a computerized solution that ensures secure data storage, quick access, and accurate management of information related to books, students, and library transactions.

The system consists of two main modules: Admin and Student. The Admin module provides full control over the system, allowing the administrator to manage the book catalog, track physical book copies, issue and return books, manage fines, generate analytical reports, and create student library ID cards. The Student module enables registered students to browse the library catalog, view their issued books, check outstanding fines, submit book requests to the administrator, and access the E-Library to read uploaded PDF books online.

Key advancements in this version of the system include an E-Library Module allowing students to read PDF books directly in the browser, a formal Book Request System through which students can request physical books and administrators can approve or reject requests, Unique Copy Tracking via a dedicated book copies table that assigns a distinct code to every physical copy (e.g., ISBN-123-1), Graphical Analytics powered by Chart.js for visualizing top-borrowed books and category-wise data, the ability to generate and print PDF reports for Issue History, Overdue Books, and Top Books, ID Card Generation for students, and a strict fine calculation system enforcing NRS 2 per day for overdue books.

The system is developed using modern web technologies including PHP and MySQL to ensure reliability, security, and ease of use. By automating library operations, it reduces paperwork, minimizes errors, and provides an efficient and user-friendly platform for both administrators and students.

Key words: *Book Cataloging, Circulation Management, Member Registration, Fine Management, Inventory Tracking, Reservation System,, PHP, MySQL Database.*

Acknowledgement

On this great occasion of the successful completion of our project, we would like to express our sincere gratitude to our Supervisor Phul Babu jha for his valuable guidance and continuous support throughout the development of this project.

We would also like to extend our heartfelt thanks to our Coordinator Naw Raj Joshi for providing us with this wonderful opportunity to work on the "Library Management System". Their support, encouragement, and insightful suggestions have been instrumental in the successful completion of this project.

Without their valuable guidance, this project would not have been possible. Their expertise and recommendations have helped us in various phases, from planning and design to implementation and testing. We truly appreciate their contribution and assistance.

Lastly, we assure that this project has been developed with our own efforts, and it is an original work.

Binita Kumari Khatri
Manisha Sapkota

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LIST OF ABBREVIATION

AI	Artificial Intelligence
CSS	Cascading Style Sheets
DFD	Data Flow Diagram
ER	Entity Relationship
HTML	Hypertext Markup Language
LMS	Library Management System
MYSQL	My Structured Query Language
PHP	Hypertext Preprocessor
RFID	Radio Frequency Identification

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Chapter 1 :Introduction

1.1 Introduction

A library plays a vital role in promoting knowledge, education, and access to information by providing books and academic resources to students and researchers. However, traditional manual library management methods are slow, error-prone, and inefficient.

The **Library Management System (LMS)** is developed to automate and digitize all library operations. It helps manage book records, track borrowing and returns, handle fines, and provide digital access to resources. This reduces paperwork, minimizes errors, and ensures accurate and quick access to information.

This upgraded system includes advanced features such as an **E-Library Module** for reading PDF books online, a **Book Request System** for requesting physical books, and **Unique Copy Tracking** for managing individual book copies. It also provides tools like graphical analytics, PDF reports, and ID card generation for better administration.

Overall, the LMS improves efficiency, organization, and accessibility, creating a modern and effective library system for both students and administrators.

1.2 Problem Statement

The existing approach to library management in many academic institutions is largely manual, making the process slow, inefficient, and prone to errors. Book records, borrowing transactions, fine calculations, and inventory tracking are often maintained in paper registers or basic spreadsheets. This makes it difficult to retrieve accurate information quickly, especially when students need to check book availability or administrators need to generate usage reports.

Due to the absence of a centralized digital system, administrators face challenges in tracking the status of individual physical copies of a book, managing student requests formally, and providing students with digital access to reading materials. Manual fine calculations are inconsistent and error-prone, and there is no reliable mechanism for producing formatted reports or library ID cards.

Furthermore, students have no self-service means to request books, browse digital resources, or check their borrowing history and outstanding fines without physically visiting the library. This lack of a student-facing digital portal reduces the overall efficiency and user satisfaction of library services.

Therefore, there is a clear need for a secure, reliable, and feature-rich Library Management System that can efficiently manage book catalogs, unique physical copy tracking, student requests, digital content delivery, automated fine calculation, graphical analytics, and administrative reporting.

1.3 Objectives

- To automate and manage library operations such as book cataloging, borrowing, returning, and physical copy tracking through a web-based system.
- To provide digital access to resources through an E-Library module and enable efficient request handling for physical books.

1.4 Scopes and Limitation

Scope

The Library Management System is designed to manage and automate the major activities of a library through an online platform. The scope of the system includes:

- **Book Catalog and Unique Copy Management:** Registration and management of all book titles and their individual physical copies, with each copy assigned a unique system-generated code (e.g., ISBN-123-1).
- **E-Library Module:** Uploading of PDF files by administrators and online reading of digital books by students through a dedicated E-Library dashboard.
- **Book Request System:** Submission of physical book requests by students and management (approve or reject) of these requests by administrators.
- **Book Borrowing and Returning:** Processing of copy-level book issuance and return transactions with automatic inventory updates.
- **Fine and Penalty Management:** Automatic calculation of overdue fines at a strict rate of NRS 2 per day and recording of fine amounts in the database.
- **Graphical Analytics and PDF Reports:** Admin dashboard charts using Chart.js and generation of printable PDF reports for Issue History, Overdue Books, and Top Books.

- **ID Card Generation:** Generation and printing of a customized library identity card for registered students by the administrator.

Limitations

Despite its advantages, the system has some limitations:

- **Dependence on Internet and Technology:** The system requires a stable internet connection and basic computer literacy to operate, which may limit accessibility in certain environments.
- **Offline Access to E-Library:** Students cannot download or access E-Library PDFs offline; an active internet connection is required.
- **Limited to Registered Users:** Only registered students and the administrator can access system services.
- **Data Accuracy Depends on User Input:** Incorrect information entered by users or administrators may result in inaccurate records.
- **Security and Privacy Risks:** Like any online system, there may be risks related to data security if not properly maintained and updated.

1.5 Report Organization

- **Chapter 1 Introduction**

  This chapter introduces the **Library Management System (LMS)** project. It covers subtopics like the Introduction, Problem Statement, Objectives, and Limitations, which explain the initial steps and motivation for building this software. It highlights the transition from traditional, paper-based library tracking to a fully automated digital system that manages both physical book circulation and E-book reading.
- **Chapter 2 Background Study and Literature Review**

  This chapter focuses on the Background Study and Literature Review of existing library management methods. It discusses the advantages and disadvantages of current manual and basic digital systems, with a focus on the pressing need for a comprehensive, student-centric platform. It emphasizes the necessity for modern features like automated fine calculation, digital PDF access (E-Library), and real-time student notifications.
- **Chapter 3 System Design**

  This chapter details the System Analysis and Design. It encompasses both Requirement Analysis and Feasibility Analysis. System analysis is the process of examining the library's workflow to ensure the information system is properly structured and modeled.

Requirement Analysis addresses both functional requirements (e.g., book issuing, return tracking, fine calculation, role-based dashboards) and non-functional requirements (e.g., system security, responsive UI). Furthermore, Feasibility Analysis addresses the technical, operational, and economic viability of the project.

- **Chapter 4 Implementation and Testing**

  This chapter explains how the system was set up, coded, tested, and debugged. The Implementation section details the technologies used (such as PHP, MySQL, CSS, and Chart.js) and the specifics of the implemented modules (Authentication, Admin/Librarian, Member, Book Management, and Circulation/Fines). The Testing section covers unit tests of critical components—such as login validation, automated fine calculations, and book request approvals—as well as comprehensive system testing to ensure the entire platform operates smoothly and securely.

- **Chapter 5 Conclusion and Future Recommendations**

  In Chapter five, the Conclusion, Lessons Learnt, and Future Enhancements are briefly explained. In the Conclusion, we deduce the overall success of the project in meeting its objectives of digitizing library operations. In Lessons Learnt, we reflect on the technical skills acquired during development, such as database relationship management, secure password hashing, and dynamic data visualization. Finally, Future Enhancements lists functionalities that could be added in later versions, such as QR/Barcode scanning for books, SMS gateway integration for overdue alerts, and an online payment gateway for library fines.

Chapter 2 :Background Study and Literature Review

2.1 Background Study

Libraries are a fundamental pillar of academic institutions, providing students, faculty, and researchers with organized access to books, references, and learning materials. Effective library management is essential for ensuring the availability of resources, maintaining accurate records, and providing a satisfactory experience for library users. However, traditional manual library management methods, such as maintaining borrowing registers, paper-based catalogs, and hand-calculated fine records, have consistently proven inefficient, time-consuming, and prone to human error.

In traditional systems, tracking the availability of individual book copies, managing borrowing queues, calculating fines accurately, and providing any form of digital resource access are virtually impossible without significant administrative effort. These manual processes hinder scalability, prevent data-driven decision-making, and create friction between library staff and users.

With the advancement of information technology and internet services, web-based management systems have become an effective solution for handling large volumes of institutional data accurately and efficiently. A digital Library Management System uses an online platform to store book catalog information, manage physical copy inventory, process borrowing and returning transactions, deliver digital content, and generate administrative reports. It improves transparency, reduces paperwork, minimizes human error, and ensures accurate and timely access to library information.

The development of a comprehensive Library Management System is therefore essential to modernize library operations in academic institutions. The system documented in this report connects administrators and students through a single digital platform, making the process of book borrowing, request management, digital reading, and library administration more organized, reliable, and effective.

2.2 Literature Review

The literature review examines existing studies, systems, and technologies related to library management to understand current practices and identify the gaps that the proposed Library Management System (LMS) aims to address. Several researchers and institutions have

emphasized the importance of automated library management for improving academic services.

Traditional manual systems are widely documented as being slow, error-prone, and difficult to scale, particularly in educational institutions with large student populations and growing book collections [1]. The introduction of computerized systems in the late 20th century began addressing these issues by automating core processes such as cataloging, book circulation, and record management.

Previous web-based library management systems have demonstrated that digital platforms reduce paperwork, improve the accuracy of borrowing records, and speed up book issuance and return processes [2]. Research has further highlighted the growing importance of providing digital content access through library systems, enabling students to utilize reading materials remotely without requiring a physical copy. Studies focusing on the Nepali educational context have specifically noted that the adoption of modern, web-based LMS in college libraries significantly improves resource utilization, system efficiency, and overall student satisfaction [3].

However, existing systems still face several operational challenges. Many basic systems track books only by total quantity, lacking the copy-level tracking granularity necessary for precise inventory management. Furthermore, formal student request systems are often absent, requiring students to request books informally at the front desk. Analytics and reporting capabilities in simpler systems are also heavily limited, preventing administrators from making data-informed decisions about the library's collection. These gaps highlight a distinct need for a more comprehensive, feature-rich platform.

The proposed system is designed by critically considering the strengths and limitations of these previous iterations. It aims to provide a complete solution that improves data management through unique copy tracking, enhances digital accessibility via an integrated E-Library module, formalizes student-administrator interactions through a dedicated Book Request System, and empowers administrators through graphical analytics, automated PDF reporting, and integrated ID card generation.

Chapter 3 :System Analysis and Design

3.1 System Analysis

Agile methodology is going to be used while building the system. It facilitates risk management, continuous improvement, and user-centric design. This project requires specific documentation, accommodates various iterative changes, delivers frequent demonstrable outputs, and uses well-understood technologies. Therefore, Agile methodology is most appropriate for this system.

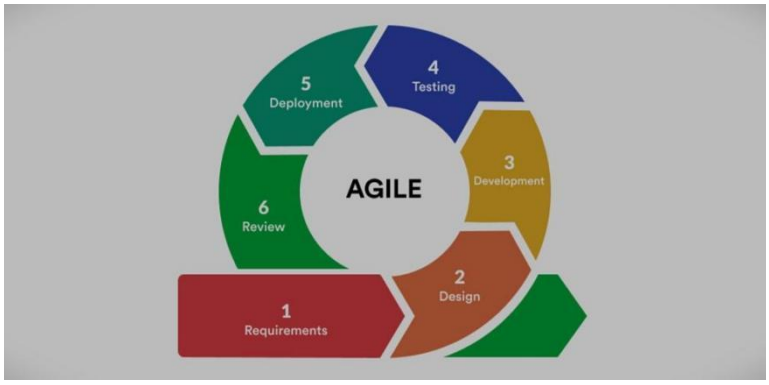

Figure 3.1Agile Methodology

3.1.1 Requirement Analysis

Requirement analysis is done while developing a system and before implementing it. It is necessary to analyze the whole system requirement. There are functional and nonfunctional requirements to be considered while determining the requirements of the system which outline what the system should do and the quality standards it must adhere to:

Functional Requirements

Functional requirements outline the specific tasks that the system must perform to manage library effectively.

- **Book Tracking and Management:** Track real-time availability of books and automatically update quantity during issue and return.
- **User Management and Access Control:** Manage user roles (admin, student) and monitor borrowing history, fines, and account status.
- **Book Borrowing and Returning:** Handle issuing and returning of books with automatic date tracking and notifications.
- **Management and Analytics:** Monitor book circulation and user activities using analytics.
- **Fine and Penalty Management:** Automatically calculate overdue fines and notify users.

- **E-Book Uploading:** Allow uploading and access of digital books.

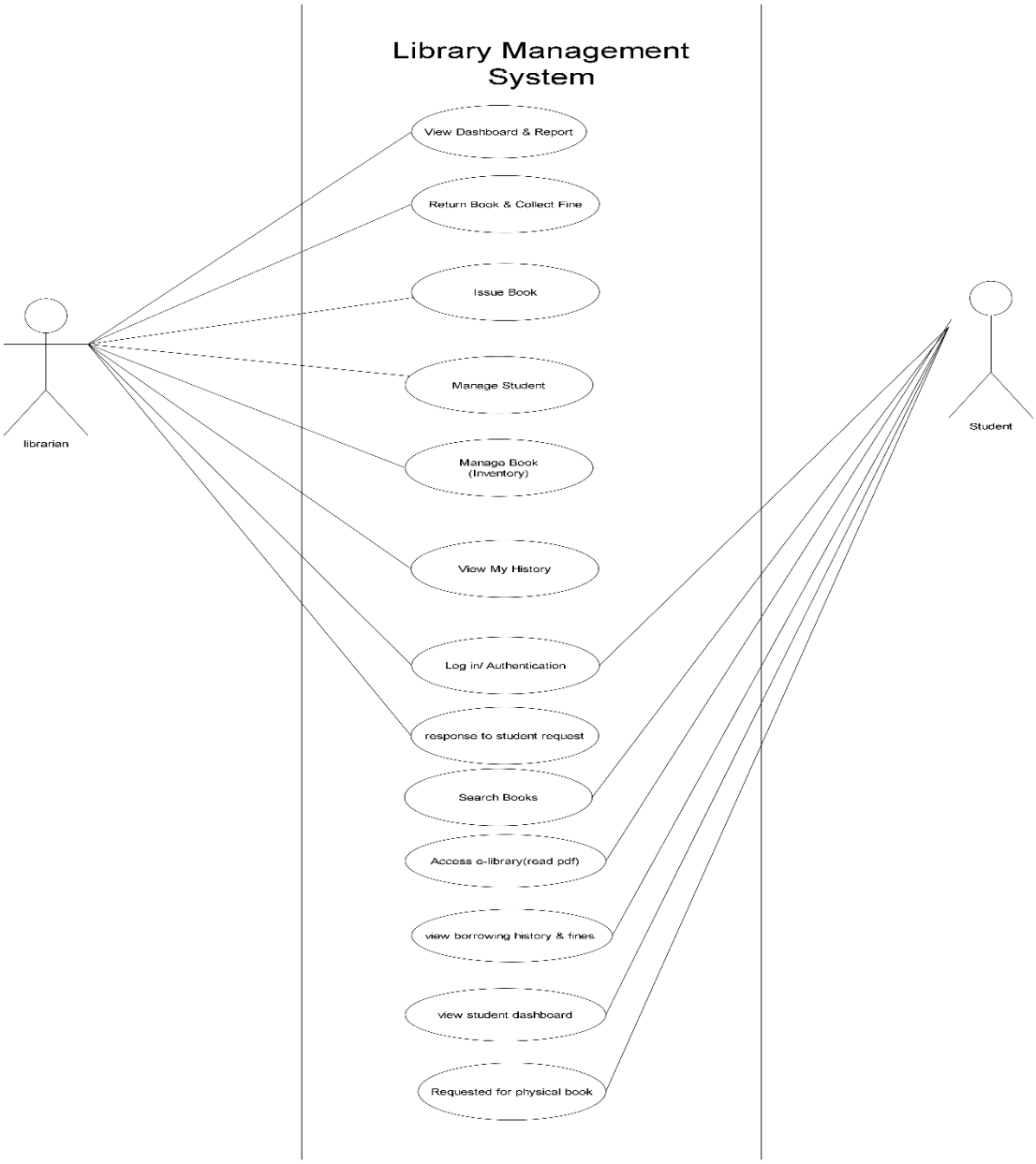

Figure 3.2 Use-Case Diagram

Non-functional Requirements

Non-functional requirements describe how the system should perform:

- **Security:** Only authorized users can access their respective modules. PHP session-based authentication is enforced throughout the system.
- **Reliability:** The system must provide accurate and consistent information, particularly for fine calculations and inventory counts.

- **Usability:** The interface must be user-friendly and easy to navigate for both administrators and students, including features such as the E-Library reader and book request workflow.
- **Performance:** The system must respond quickly to user requests, including book searches, PDF loading, and chart rendering.
- **Availability:** The system must be accessible online whenever needed.
- **Data Integrity:** The system must ensure correct and consistent storage and retrieval of data, particularly across related tables such as books, book_copies, and book_requests.

iii. Technical Requirements:

Technical requirements specify the hardware, software, and system environment needed for the system to function.

- **Hardware Requirements:** The system must run on standard server hardware or a local XAMPP/WAMP server environment, with provisions for adequate storage to accommodate the PDF library collection.
- **Software Requirements:** The system is developed using PHP as the server-side language and MySQL as the database management system. The frontend uses HTML, CSS, and JavaScript with Chart.js for data visualization and a server-side PDF generation library for report output.
- **Platform Compatibility:** The system must be accessible via modern web browsers such as Google Chrome and Mozilla Firefox on desktop and mobile devices.

3.1.2 Feasibility Analysis

Feasibility Analysis is the process of evaluating whether the proposed Library Management System is practical and beneficial for implementation. It examines different aspects to determine if the system can be developed successfully within the available resources and constraints.

- **Technical Feasibility:** The LMS is technically feasible because it is built using widely available and well-supported technologies: PHP, MySQL, HTML, CSS, JavaScript, and Chart.js. These technologies are mature, open-source, and do not require expensive infrastructure. The integration of Chart.js for client-side data visualization and a PHP-based PDF generation library are well within the capabilities of the chosen technology stack. File system management for PDF uploads is fully supported by PHP's native file handling

functions. The expanded database schema, including the new book_copies and book_requests tables, is fully compatible with MySQL's relational model.

- **Economic Feasibility:** The proposed system is economically feasible because the development cost is low compared to the benefits it provides. All core components (PHP, MySQL, Chart.js) are open-source, eliminating software licensing costs. The system reduces administrative workload through automation, eliminates the need for separate ID card printing processes, and provides data-driven insights through analytics — all of which contribute to cost savings and improved resource allocation in the institution.

- **Operational Feasibility:** The LMS is operationally feasible because it is designed to be user-friendly for both the administrator and students. The E-Library module reduces physical borrowing load. The automated Book Request workflow reduces the need for manual inquiries at the front desk. The automated fine calculation at NRS 2 per day ensures consistent enforcement without administrative effort. The system integrates naturally into the daily operations of an academic library without requiring significant retraining of staff or students.

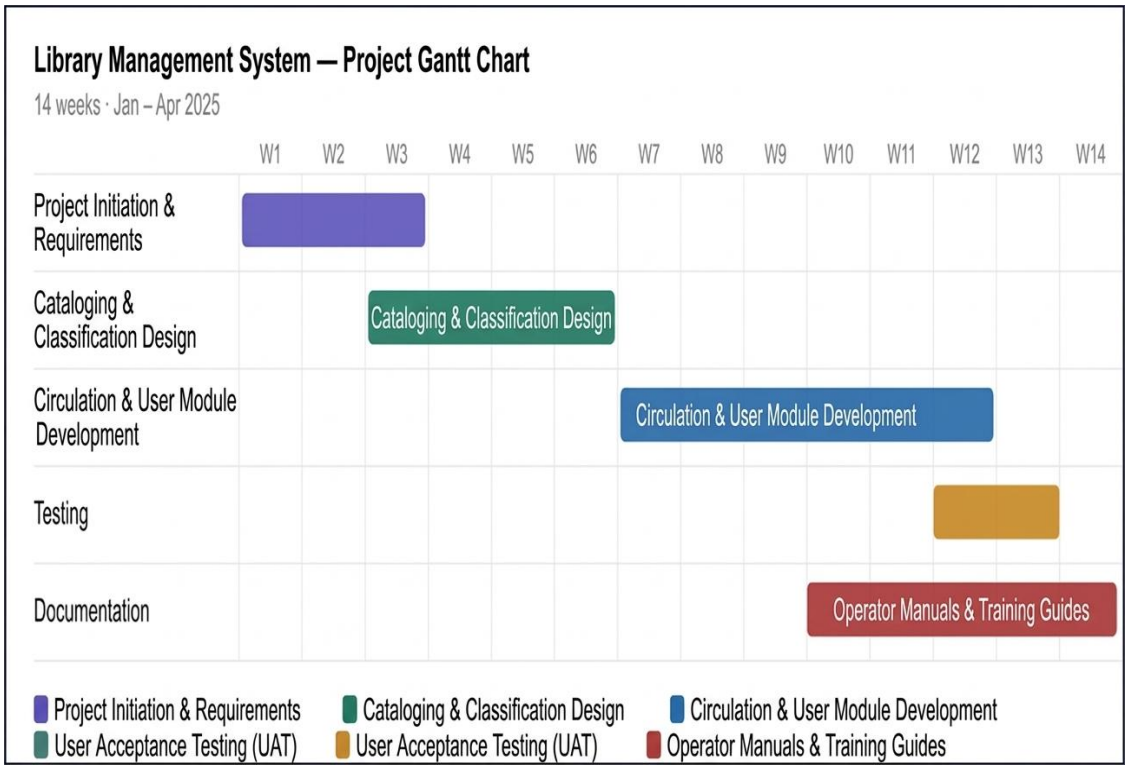

Figure 3.3 Gannt Chart

3.1.3 Data Modelling (ER-Diagram)

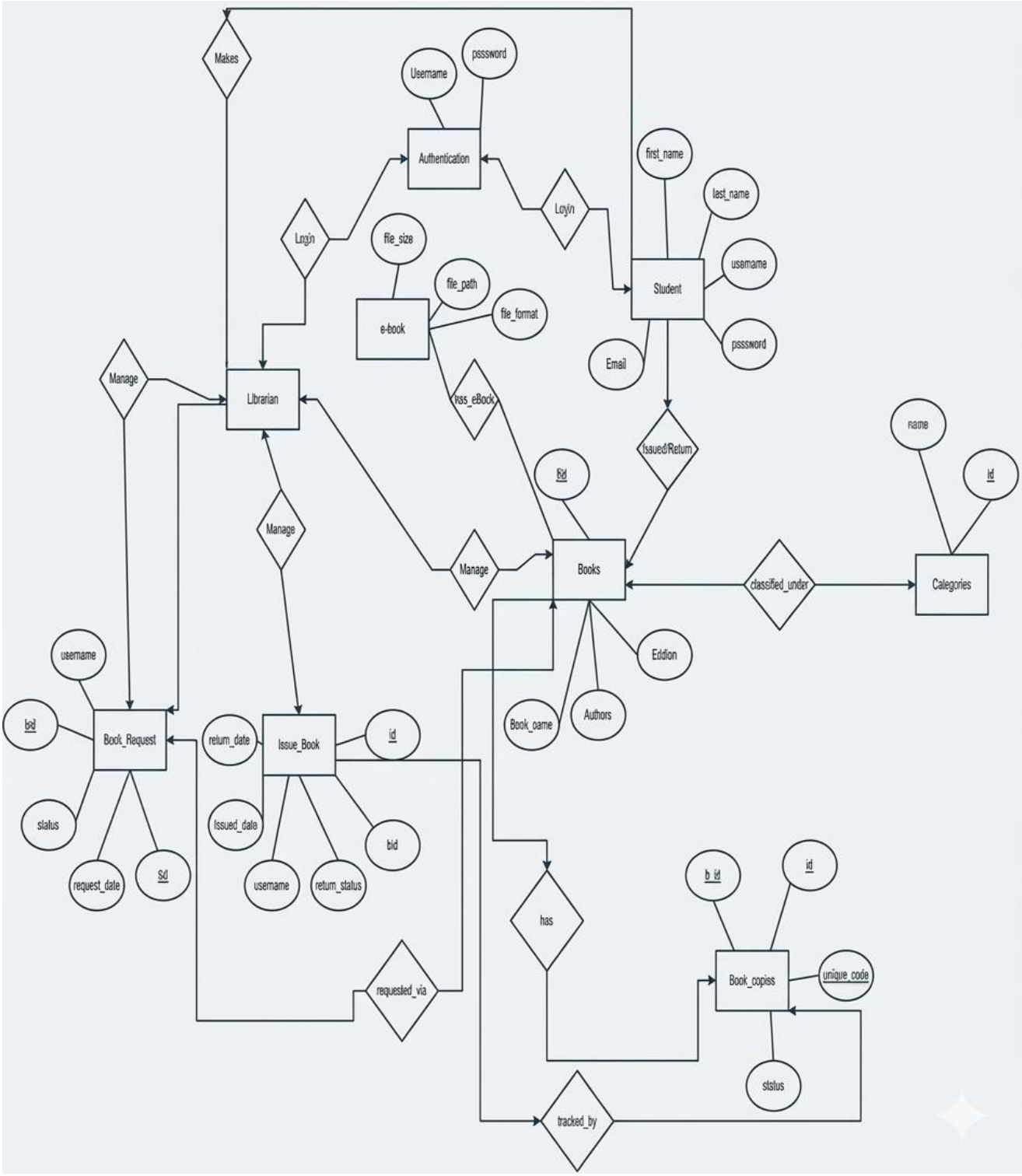

Figure 3.4 ER-Diagram

3.1.4 Process Modelling (DFD)

DFD Level 0

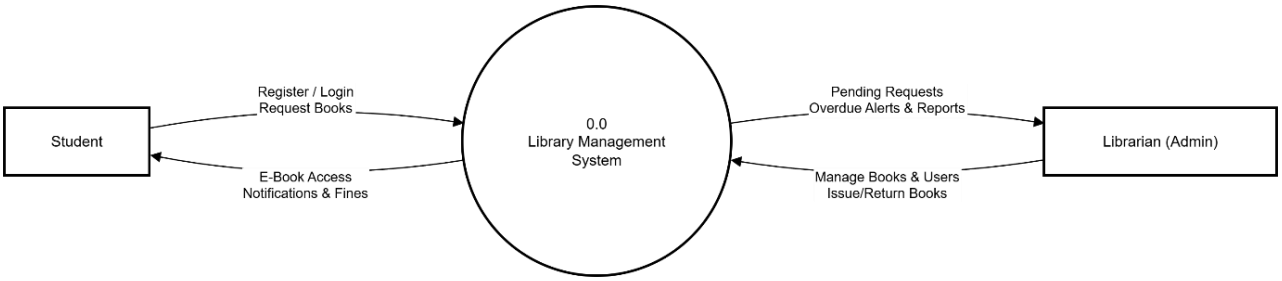

Figure 3.5 Level 0 DFD of Library Management System

DFD Level 1

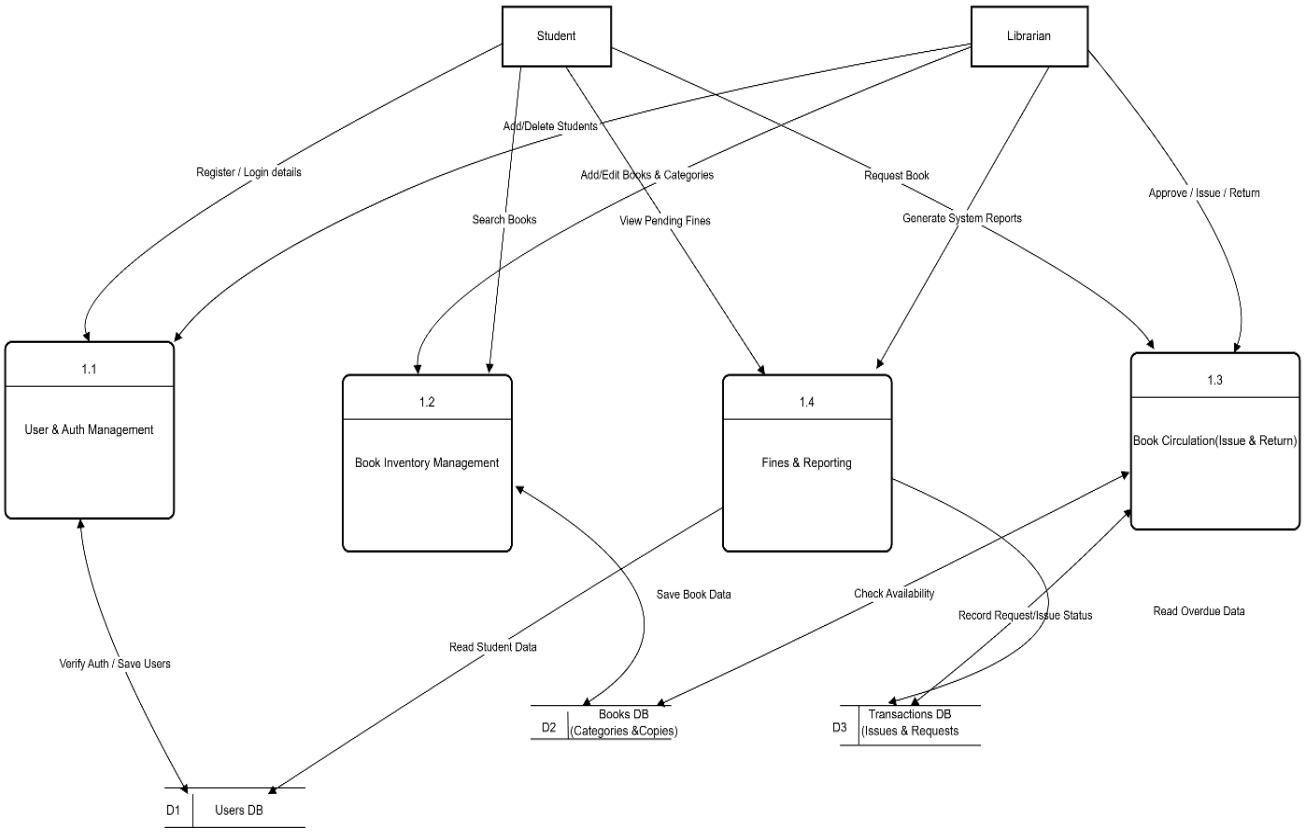

Figure 3.6 Level 1DFD of Library Management System

3.2 System Design

3.2.1 Architectural Design

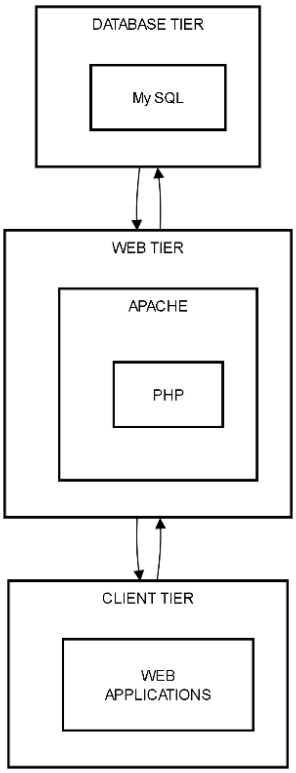

Figure 3.7 Architectural Design of Library Management System

3.2.2 Database Schema Design

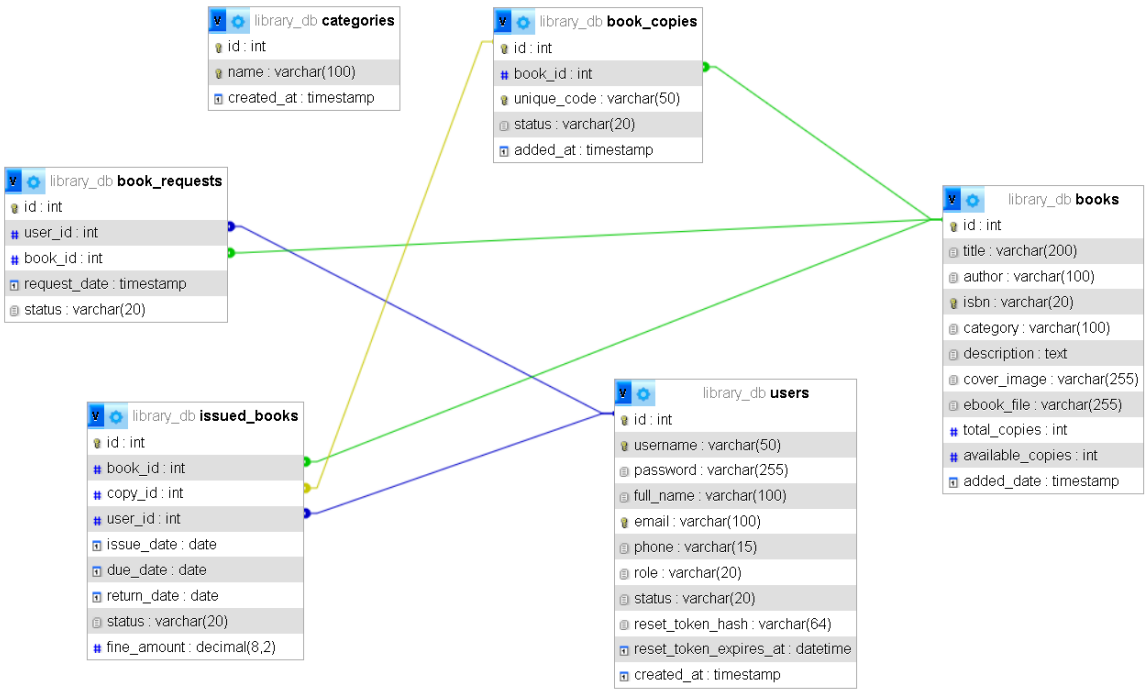

Figure 3.8 Database Schema Design of library Management System

3.2.3 Interface Design

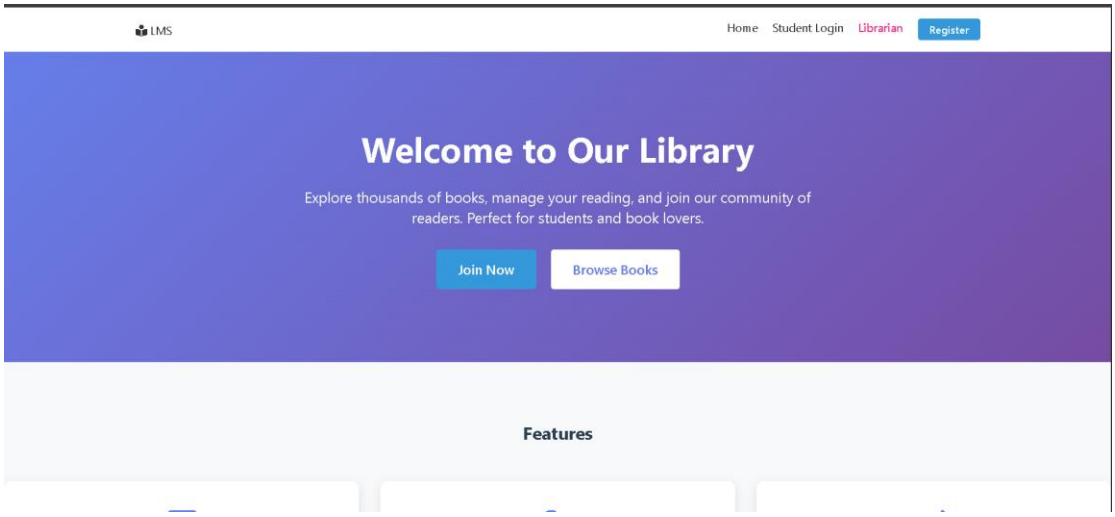

Figure 3.9 Landing page

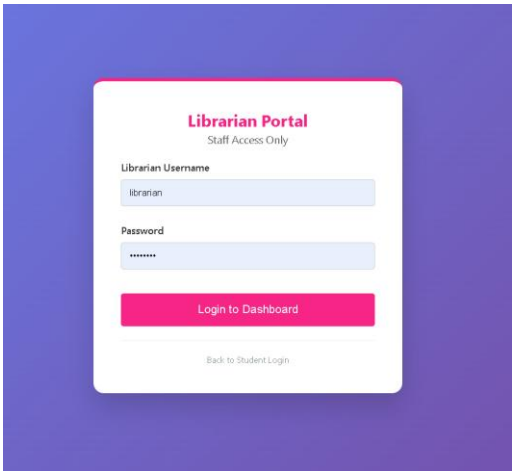

Figure 3.10 Login page for Librarian portal

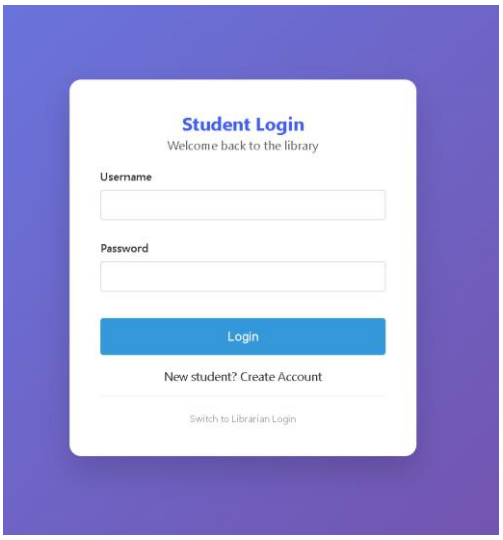

Figure 3.11 Login page for student

3.2.4 Physical Data Flow Diagram

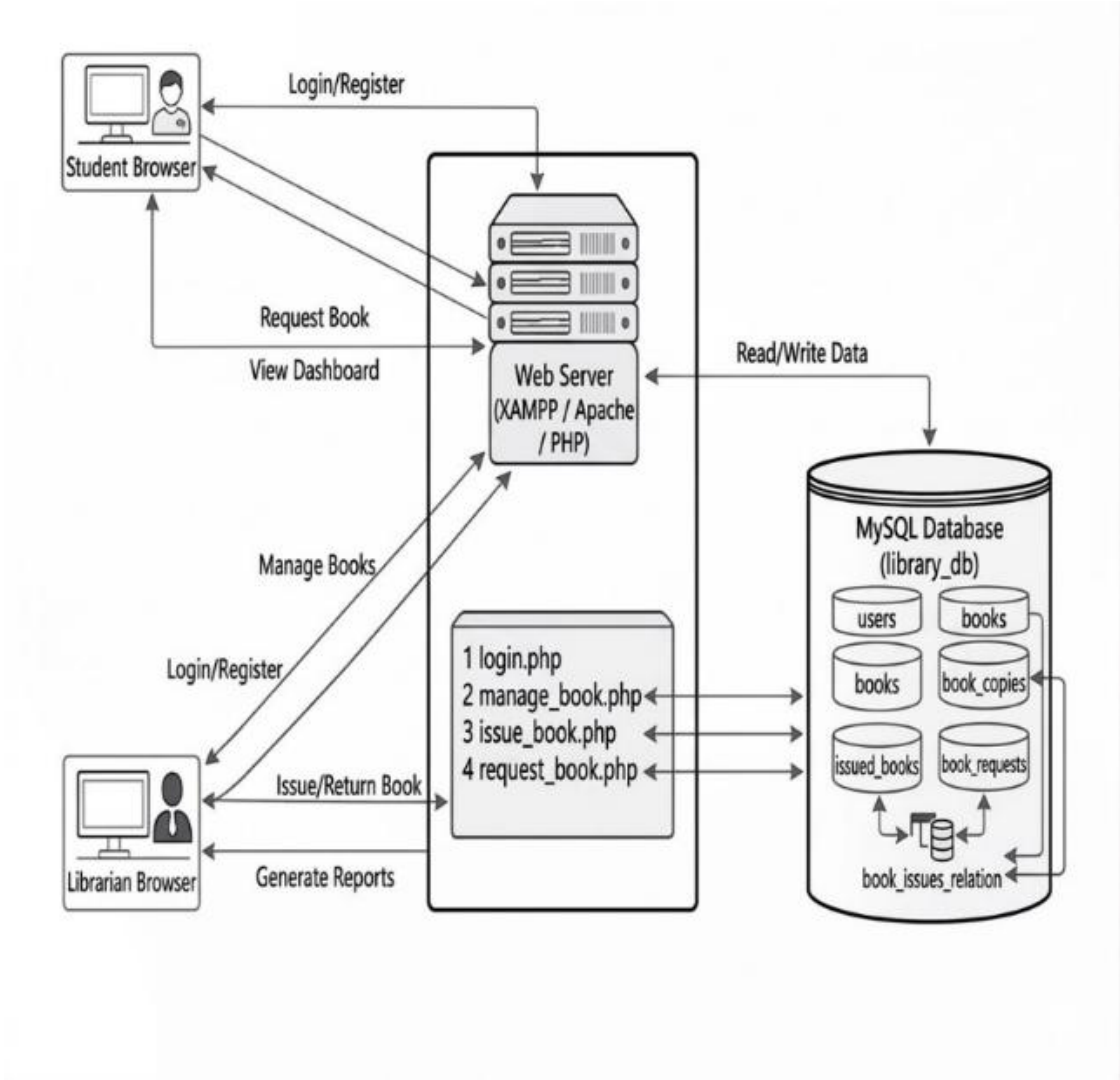

Figure 3.12 Physical DFD diagram

Chapter 4 :Implementation and Testing

4.1 Implementation

During the implementation phase Library management System , a combination of programming language, and development tools were utilized to create a comprehensive and user-friendly system.

4.1.1 Tools Used

Table 4.1 Tools Used

Frontend Technologies	HTML, CSS, JavaScript,
Backend Technologies	PHP
Database Technologies	MySQL
Development Tools:	Version Control : Git Text Editor: VS Code Diagram: Draw.io and DrawSQL

4.1.2 Implementation Details of Module

The proposed Library Management System is composed of several interconnected modules such as the Authentication Module, Member (Student) Module, Admin (Librarian) Module, Book Management Module, Circulation (Issue/Return) Module, and Reporting Module. Below is an overview of each module:

- **Authentication (Login/Registration) Module:**

  This module securely handles user access. Students can create an account by filling out a registration form with details like their full name, email, username, and password. Passwords are encrypted using secure hashing algorithms. The system features separate, dedicated login portals for Students and Librarians to ensure secure role-based access control.

- **Librarian Module:**

  In this module, the librarian logs in to access a comprehensive administrative dashboard. The dashboard displays real-time statistics and graphical charts (e.g., most popular books, library composition). The admin manages the entire ecosystem, including student accounts, book inventories, and categories. The librarian can also generate and print official Library ID cards for registered students.

- **Student Module:**

Registered students use this module to interact with the library. Upon logging in, they access a personalized dashboard showing their currently borrowed books, overdue status, and pending fines. Students can browse the library catalog, filter books by category, request physical books, and directly read uploaded E-books (PDFs) online. A dedicated notifications tab provides a timeline of approved requests, book issues, returns, and overdue alerts.

- **Book Management Module:**

  This module allows the librarian to add, update, and delete books in the system. When adding a book, the admin provides the title, author, ISBN, category, and total copies. The system automatically generates unique tracking codes for every physical copy. Additionally, the librarian can upload book cover images and attach digital PDF files to create an E-library collection.

- **Circulation & Request (Issue/Return) Module:**
- This module handles the lifecycle of book borrowing. Students initiate the process by sending a "Book Request." The librarian reviews these requests in the admin panel and can approve (issue) or reject them based on stock availability. The librarian can also manually issue specific physical copies to students and set due dates.
- **Fine & Return Management Module:**

  When a student returns a book, the librarian processes it through this module. If the book is returned past its due date, the system automatically calculates the pending fine (e.g., NRS 2 per day) and logs the collected amount into the system's fine history.

- **Reporting Module:**

  This module assists the librarian in tracking library activities. The admin can generate, filter by date, and print detailed reports, including Issue History, Overdue Books, and Top/Most Popular Books.

4.2 Testing

Testing is the process of determining whether the system works effectively and efficiently. Testing does not only include debugging. It also checks for quality assurance, validation and verification, reliability and availability estimation. Testing can be done at different levels.

4.2.1 Test cases for Unit Testing

The unit testing part of testing methodology is the testing of individual software modules or components that make up a web application or system.

The unit testing part of testing methodology is the testing of individual software modules or components that make up a web application or system.

Table 4.2 Login Test Case

SN	Test Case	Expected Result	Test Result
1	Enter valid username and password and click Login	Should display main dashboard for admin or respective student dashboard	Successful
2	Enter invalid username and password	Should display "Invalid details" alert	Successful

Table 4.3 Registration Test Case

SN	Test Case	Expected Result	Test Result
1	Submit empty registration form	Browser displays HTML5 validation prompts for required fields	Successful
2	Submit with password mismatch	Displays "Passwords do not match" alert	Successful
3	Submit with invalid phone number (less than 10 digits)	Displays "Please enter a valid 10-digit phone number" alert	Successful
4	Submit with existing email address	Displays "Username or Email already exists" error	Successful
5	Submit without enrollment code	Displays "Code Required" validation message	Successful
6	Submit all valid details	Registration successful; user redirected to login page	Successful

Table 4.4 Add Book Test Case

SN	Test Case	Expected Result	Test Result
1	Submit book form without selecting Category	HTML5 "Select an item in the list" prompt displayed	Successful
2	Submit book with an existing ISBN	Displays "ISBN exists" alert	Successful
3	Add new Category via the modal	New category added to dropdown and selected	Successful
4	Submit valid book with Copies = 5	Book added; 5 unique copy codes generated in book_copies table; redirect to Manage Books	Successful

Table 4.5 Book Issuance and Return

SN	Test Case	Expected Result	Test Result
1	Admin issues a book to a student (select student and book copy)	Book issued; copy status updated to 'issued' in book_copies; available count decreases	Successful
2	Student views issued book on dashboard	Issued book appears in 'My Books' with Active status and due date	Successful
3	Admin returns a book before due date	Status becomes 'returned'; no fine applied; inventory count increases	Successful

SN	Test Case	Expected Result	Test Result
4	Admin returns an overdue book	System calculates fine (overdue days × NRS 2) and displays fine amount in alert	Successful
5	Admin completes return process with fine	Fine amount recorded in issued_books table	Successful

Table 4.6 E-Library Module

SN	Test Case	Expected Result	Test Result
1	Student navigates to E-Library dashboard	E-Library page loads showing list of books with available PDFs	Successful
2	Student clicks 'Read' on a book with uploaded PDF	PDF opens and renders correctly in the browser; no borrowing transaction created	Successful
3	Student clicks 'Read' on a book without PDF	No PDF link displayed; 'Read' button not available	Successful

Table 4.7 Book Request System

SN	Test Case	Expected Result	Test Result
1	Student submits a book request	Request saved in book_requests table with status 'pending'; confirmation message shown	Successful
2	Admin views Manage Requests page	All pending requests listed with student name and book title	Successful

SN	Test Case	Expected Result	Test Result
3	Admin approves a book request	Request status updated to 'approved'; book automatically issued to student; copy status updated	Successful
4	Admin rejects a book request	Request status updated to 'rejected'; student notified on their portal	Successful

Table 4.8 Analytics, Reports, and ID Card

SN	Test Case	Expected Result	Test Result
1	Admin views the dashboard	Chart.js charts render correctly for Top Books and Category-wise data	Successful
2	Admin generates Issue History PDF report	PDF report generated and print dialog opens	Successful
3	Admin generates Overdue Books PDF report	PDF report showing all overdue books and fine amounts generated	Successful
4	Admin clicks 'ID Card' for a student	Library ID Card print view opens with student name, ID number, and library details	Successful

Table 4.9 Logout

SN	Test Case	Expected Result	Test Result
1	Click on Logout	User session destroyed and user redirected to login page	Successful

Chapter 5 :Conclusion and Future Recommendations

5.1 Outcomes

Through the development of the Library Management System (LMS), valuable lessons were learned that guided the project toward successful completion. Establishing a well-defined project scope at the outset, adopting an Agile iterative development approach for flexibility, and maintaining clear communication among team members were essential in meeting all system requirements. The development process emphasized proper requirement analysis, continuous testing at both unit and system levels, and early identification and resolution of issues, which helped ensure the reliability and efficiency of the final system.

The outcomes of the project include a fully functional and feature-rich Library Management System encompassing: a complete book catalog with copy-level unique tracking via the book_copies table, an E-Library module for online PDF reading, a formal Book Request System with admin-side approval management, automated book issuance and return processing, strict NRS 2/day overdue fine calculation, Chart.js-powered graphical analytics, PDF report generation for Issue History and Overdue Books, and Library ID Card generation for students.

Additionally, the Library Management System contributes to improving the coordination between administrators and students, facilitating faster access to both physical and digital library resources, enhancing record accuracy, and promoting organized and transparent library operations within the academic institution.

5.2 Conclusion

The Library Management System was developed to provide an efficient, reliable, and user-friendly solution for managing all activities of an academic library. The system successfully achieves its main goal of connecting students and administrators through an online platform that streamlines book cataloging, unique copy management, borrowing, returning, digital content access, and administrative reporting. By computerizing manual library operations, the system reduces paperwork, saves time, minimizes errors, and improves the accuracy of information management.

The project demonstrates how technology can be applied effectively to institutional management challenges. Features such as the E-Library module, the Book Request System, unique copy tracking, Chart.js analytics, and PDF report generation ensure

transparency, data-driven administration, and enhanced student access to library resources. The system also improves coordination between administrators and students and supports better decision-making for library resource management.

Although the system meets all its stated objectives, there is still scope for future enhancements such as mobile application support, real-time notifications, barcode or RFID integration, cloud-based PDF storage, and advanced security features. Overall, the Library Management System proves to be a practical and beneficial academic project that contributes meaningfully to improving library services in educational institutions.

5.3 Future Recommendations

While the Library Management System has successfully achieved its primary objectives, several areas present opportunities for further improvement in functionality, efficiency, and user experience.

- **Book Reservation and Waitlist System:** Extend the current Book Request System with a waitlist mechanism, automatically notifying students and fulfilling their requests when a currently issued copy is returned.
- **Automation Technologies:** Integrate barcode scanning or RFID technology to automate the physical book issuance and return processes, linking physical copy scans directly to the book_copies table records.
- **Demand Prediction:** Implement AI-based analytics to forecast book demand based on historical borrowing data and seasonal patterns, helping administrators make informed acquisition decisions.
- **Offline Functionality:** Provide limited offline features through a Progressive Web Application (PWA) approach, allowing students to access cached E-Library content in areas with limited internet connectivity.

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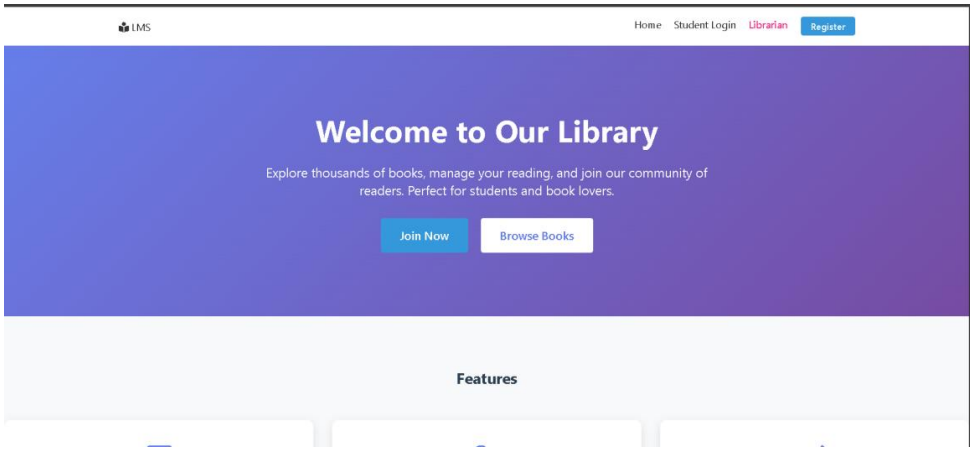
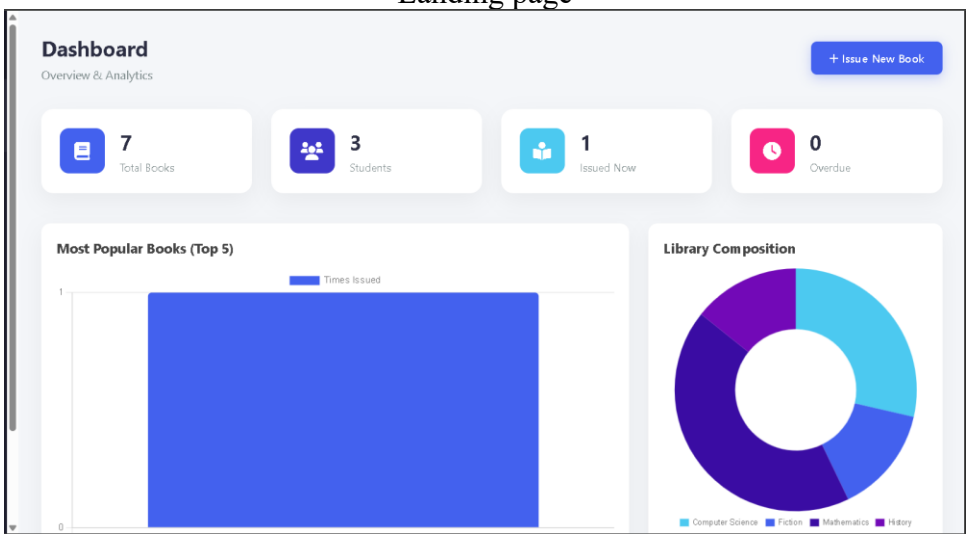
[2] Devi, P. (2022). Functionalities and advantages of web-based and open-source integrated library systems in modern education. *International Journal of Information Technology and Library Science, 14*(3), 112-125.

[3] Neupane, R. (2023). Usability evaluation of integrated library systems and automation in Nepalese institutional libraries. *Asian Journal of Information Science and Technology, 9*(1), 77-89

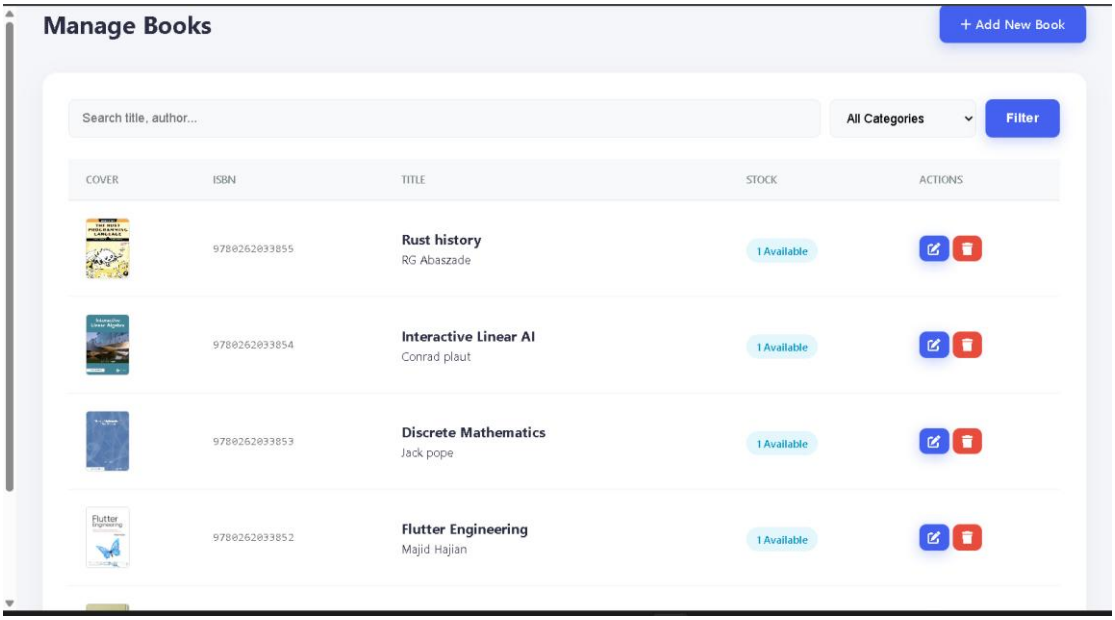
APPENDIX



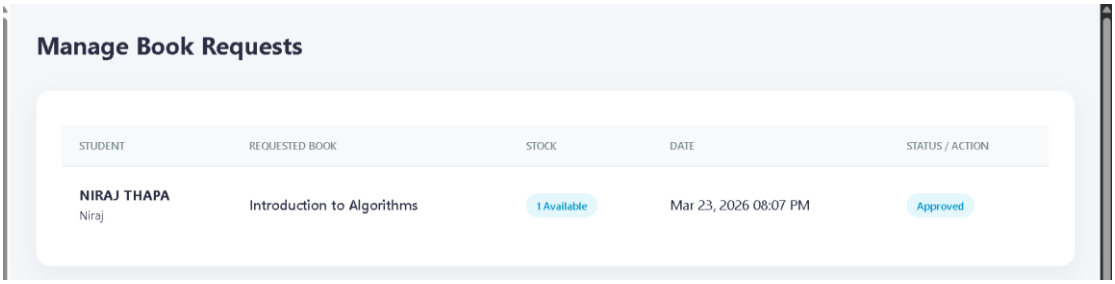
Landing page



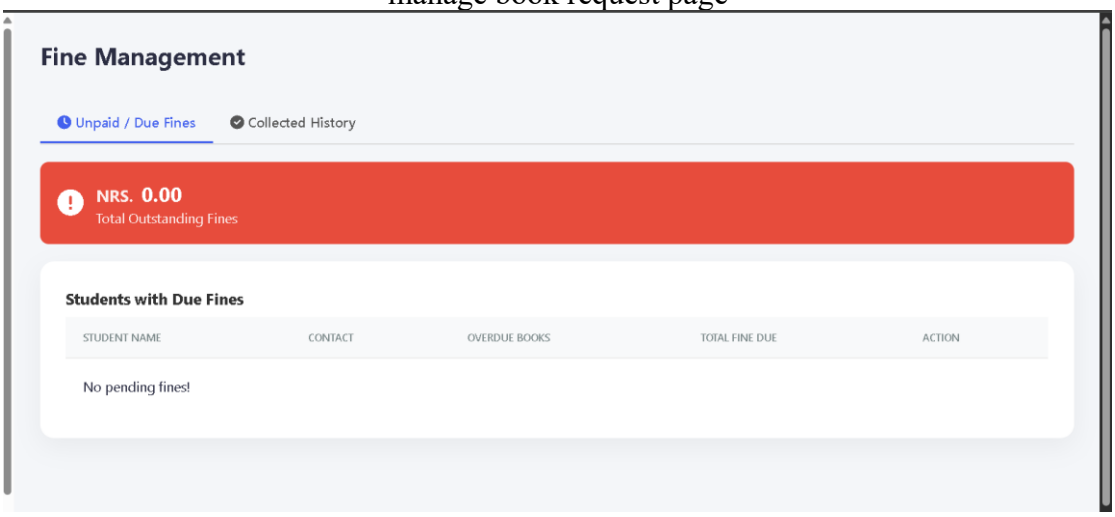
Admin dashboard



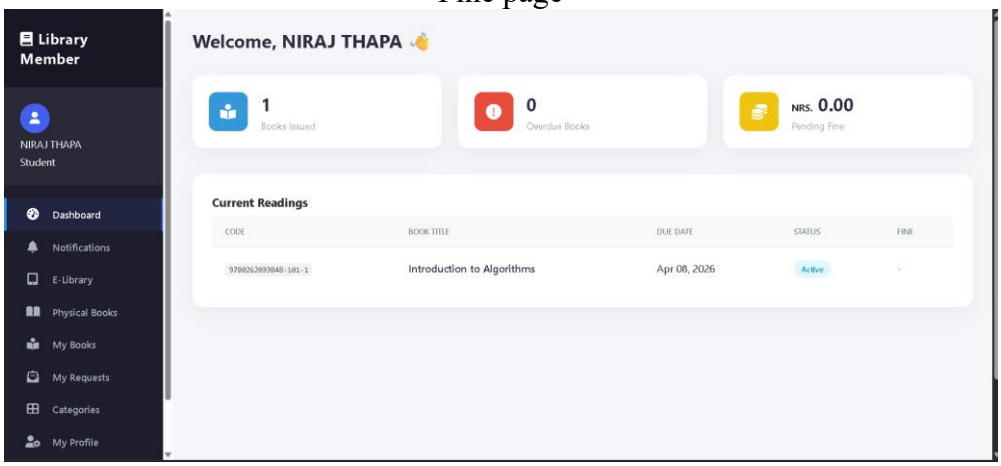
Manage Book page



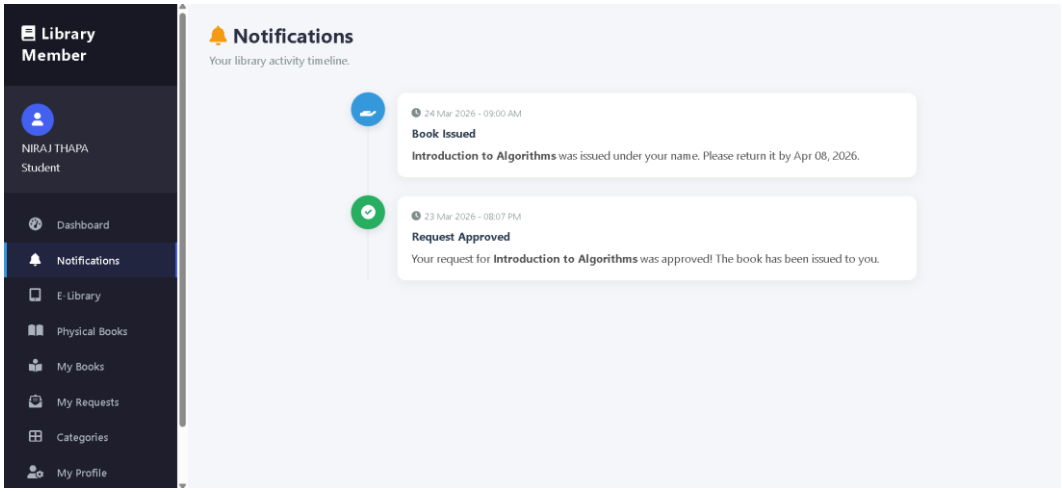
manage book request page



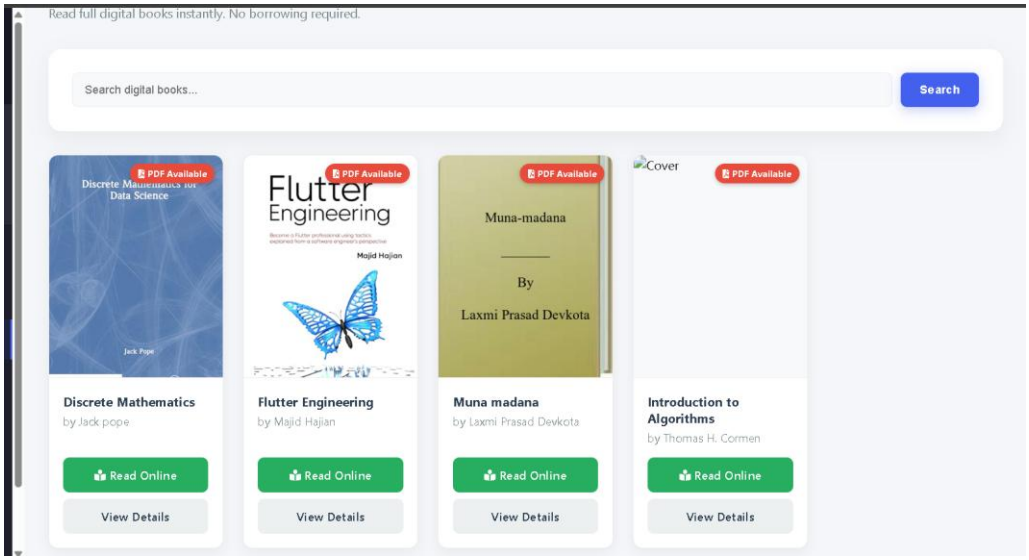
Fine page



Student dashboard



Notification page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